

On the Multilinearity of the Determinant

Clara Purdue was asking about the multilinearity of the determinant, so I thought I would say a few things here. (Although this is covered to some extent in Chapter 2 of Leon, I may have neglected to cover this in lecture.)

The *multilinearity* of the determinant can be summarized quickly as the following property of $\det$:

Lemma 1 *Let F be any field and $A \in F^{n \times n}$. Suppose, for all $i \in \{1, \dots, n\}$, that the i^{th} row of A is α_i and the i^{th} column of A is a_i . Also let $a'_1 \in F^{n \times 1}$ and $\alpha'_1 \in F^{1 \times n}$ and $\gamma, \delta \in F$. Then*

$$\det[\gamma a_1 + \delta a'_1, a_2, \dots, a_n] = \gamma \det[a_1, a_2, \dots, a_n] + \delta \det[a'_1, a_2, \dots, a_n]$$

and

$$\det \begin{bmatrix} \gamma \alpha_1 + \delta \alpha'_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} = \gamma \det \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix} + \delta \det \begin{bmatrix} \alpha'_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{bmatrix}.$$

Thanks to the alternating property of the determinant with respect to swapping rows and columns, a similar statement can be made for *any* fixed column (resp. row) that is a linear combination of other column (resp. row) vectors.

As a simple example, observe that

$$\det \begin{bmatrix} x+1 & 1 \\ 1 & x+1 \end{bmatrix} = x \det \begin{bmatrix} 1 & 1 \\ 0 & x+1 \end{bmatrix} + \det \begin{bmatrix} 1 & 1 \\ 1 & x+1 \end{bmatrix}$$

if we apply multilinearity to the first column of the left-hand matrix above. If we then apply multilinearity to the second columns of *both* the matrices on the right-hand side above, then we get:

$$\begin{aligned} \det \begin{bmatrix} x+1 & 1 \\ 1 & x+1 \end{bmatrix} &= x \left(x \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right) + \left(x \det \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \det \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) \\ &= x^2 \det \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + x \det \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} + x \det \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} + \det \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ &= x^2 + 2x \end{aligned}$$

While it may be obvious that the 2×2 determinant above can be computed more efficiently other ways (e.g., via the alternating sum of products expansion), multilinearity turns out to be much more useful for certain larger, structured matrices...